Genre effects on metaphor versus literal processing in the 'maze' task

Previous work has shown that metaphors only require additional effort to process when they are not supported by *linguistic* context [1-3]. However, metaphors 'in the wild' not only appear in a linguistic context but are also produced within an extra-linguistic context or *genre* (i.e., any group of linguistic activity structured by conventions on form and content). Corpus studies have shown that metaphors vary systematically across different genres [4-6]. For instance, the metaphor "breeze of contentment" would be atypical in a newspaper context, (1), but more natural in a fiction context, (2). Nonetheless, it is unclear whether a lack of fit with the genre can inhibit metaphor processing in the same way that the lack of linguistic context can. Our study is the first to investigate whether genre can affect metaphor processing in a word-by-word reading task.

We extracted English genitive metaphors (NOUN-of-NOUN) that were determined to be typical of *fiction* (see [6]) from the Corpus of Contemporary American English (COCA, [7]) and embedded them in *news* sentences from COCA matching the topic of the metaphors (n=12). This allowed us to create a mismatch between metaphors and their genre. We then generated literal equivalents using the large language model BERT [8]. There were no differences in KenLM-generated surprisal scores [9] for the metaphors versus the literal sentences (β=-0.54, SE=0.35, p>0.05); thus, a difference in RT would not be expected based on predictability alone.

Experiment 1 established a baseline comparison between metaphors and their literal counterparts when presented *without any context* in a 'maze' reading task [10]. At each word in the sentence, participants selected between a correct and an incorrect sentence continuation, with the critical regions being the first and second noun of the genitive metaphors and their literal counterparts (e.g., "breeze/sense of contentment", Table 1d/e; R4 & R6). Reaction time (RT) and 'maze' accuracy were analyzed using linear and logistic mixed effects models. Results from 48 self-reported L1 English speakers show a significant difference in RT between the metaphorical and literal items at the first region of interest, "[breeze vs. sense]" (R4; β=143.6, SE=49.5, p=0.009), but no difference at the second region of interest, "contentment" (R6; β=58, SE=51, p=0.27) (see Figure 1). No difference in accuracy was found (all ps > 0.05).

Experiment 2 introduced a genre manipulation. Half of our participants were informed of the genre of the item (Explicit Genre group, N=37; see Table 1a), while the other half were uninformed (Implicit Genre group, N=35). To provide this implicit information, we added two context sentences from news before each trial (see Table 1b/c), presented normally (i.e., not in 'maze'). We predicted that genre would have less of an effect if intuited from the text alone (or that participants may not be able to identify it at all). If genre mismatch inhibits metaphor processing, then metaphors in this study should lead to more processing difficulty (higher RT) than literal language because they are bad fits for a news genre; literal language is not subject to the same genre norms [4-5]. Data were analyzed as in Exp. 1, with the additional fixed effect of Genre Explicitness and an interaction term. There was no significant effect of Metaphoricity or Genre Explicitness, and no interaction (all ps > 0.05) (see Figure 2). Thus, metaphors and literal sentences led to similar RT and accuracy across Explicit/Implicit conditions.

Our findings have several implications. The fact that explicitly mentioning the genre did not influence the results could suggest that participants were able to infer the genre of the passages without being explicitly informed. This would suggest that genre expectations may not affect on-line metaphor processing above and beyond the facilitatory effect of providing context. Alternatively, it is possible that our literal sentences matched the metaphors too closely in meaning, which would make them odd for the genre. These possibilities invite a broader discussion on how effects of the extra-linguistic context are to best be studied empirically [11]. Connections to EEG results on metaphor processing will also be discussed.

References

[1] Ortony et al. (1978), *Journal of verbal learning and verbal behavior* **[2]** Inhoff et al. (1984), *Memory & Cognition* **[3]** Gibbs & Gerrig (1989), *Metaphor & Symbol* **[4]** Deignan et al. (2013), *Figurative language, genre, and register* **[5]** Goatly (2011), *The language of metaphors* **[6]** Beekhuizen & Raihert (under review) **[7]** Davies (2008), *COCA* **[8]** Devlin et al. (2019), *Proceedings of NAACL* **[9]** Heafield (2011), *Proceedings of the 6th Workshop on Statistical Machine Translation* **[10]** Forster et al. (2009), *Behavior research methods* **[11]** Gibbs (2023), *Cognition*.

Examples

(1) When Hollande's hour-long speech wound to a close, the crowd intoned the national anthem. Hollande felt a **breeze of contentment** as he stepped down from the stage.

(2) Lincoln is tired of nobility. It has been years since he felt a single **breeze of contentment**. This he blames on nobility. Rectitude exhausts his every part.

Table 1. Example of a stimulus set for Experiments 1-2 (no context in Exp. 1).

		Explicit Genre	Implicit Genre
a	**Context 0**	This passage comes from the New York Times.	---
b	**Context 1**	*Many banner-waving supporters jammed the convention centre.*	
c	**Context 2**	*When Hollande's hour-long speech wound to a close, the crowd intoned the national anthem.*	
d	**Metaphor**	*Hollande* $_{R1}$*/ felt* $_{R2}$*/ a* $_{R3}$*/* ***breeze*** $_{R4}$*/* ***of*** $_{R5}$*/* ***contentment*** $_{R6}$ */ as* $_{R7}$*/ he* $_{R8}$*/ stepped* $_{R9}$*/ down* $_{R10}$*/ from* $_{R11}$*/ the* $_{R12}$*/ stage.* $_{R13}$*/*	
e	**Literal**	*Hollande* $_{R1}$*/ felt* $_{R2}$*/ a* $_{R3}$*/* ***sense*** $_{R4}$*/* ***of*** $_{R5}$*/* ***contentment*** $_{R6}$ */ as* $_{R7}$*/ he* $_{R8}$*/ stepped* $_{R9}$*/ down* $_{R10}$*/ from* $_{R11}$*/ the* $_{R12}$*/ stage.* $_{R13}$*/*	

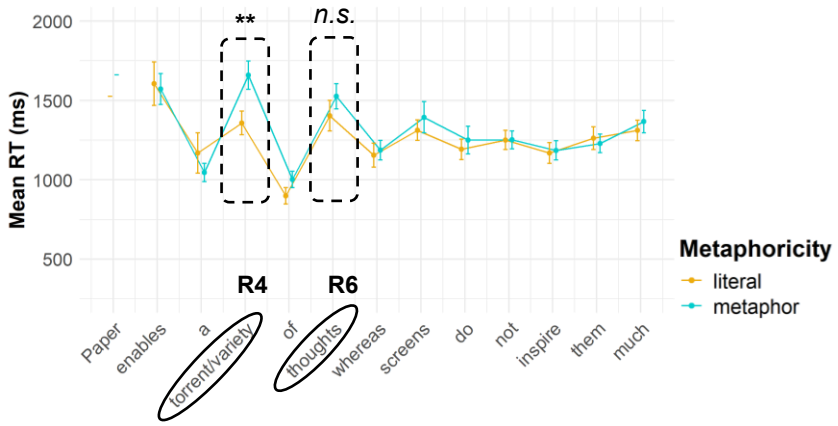

Figure 1. Experiment 1: Mean RT by sentence region and condition. There is a significant difference in RT at Region 4: metaphors are associated with a higher RT.

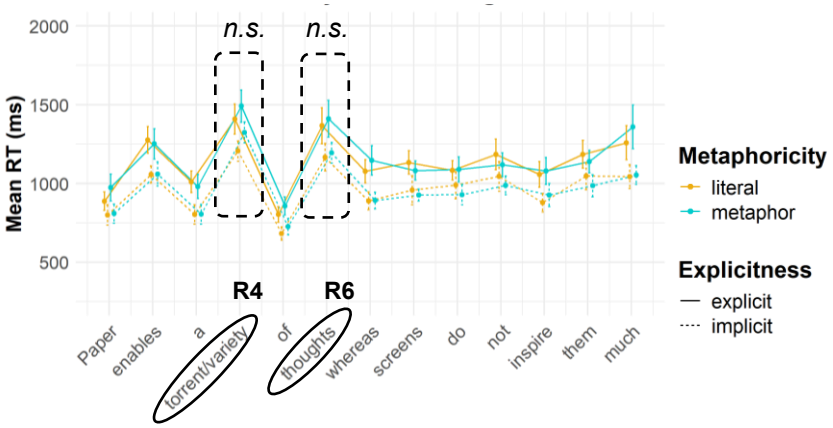

Figure 2. Experiment 2: Mean RT by sentence region and condition. There is no significant difference across conditions.